



ODP 7: Final Report - Team MANN

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Design Reflection

DESIGN FOR MANUFACTURING

Reflection

To make the pole cleaner device in large quantities, first we would need a way to produce the small parts in large quantities, and then a method for producing the whole assembly in large quantities. To produce the hinges and dispenser device out of plastic, we could use rotation molding or injection molding rather than 3D printing to make our production time and costs more efficient. This would also allow us for full freedom on our geometry. The hinges could also be redesigned and created entirely out of plastic, which would eliminate the need to cut steel rods that span the length of the device (only adding rods near the hinges). and make retaining pieces for the hinge rods.

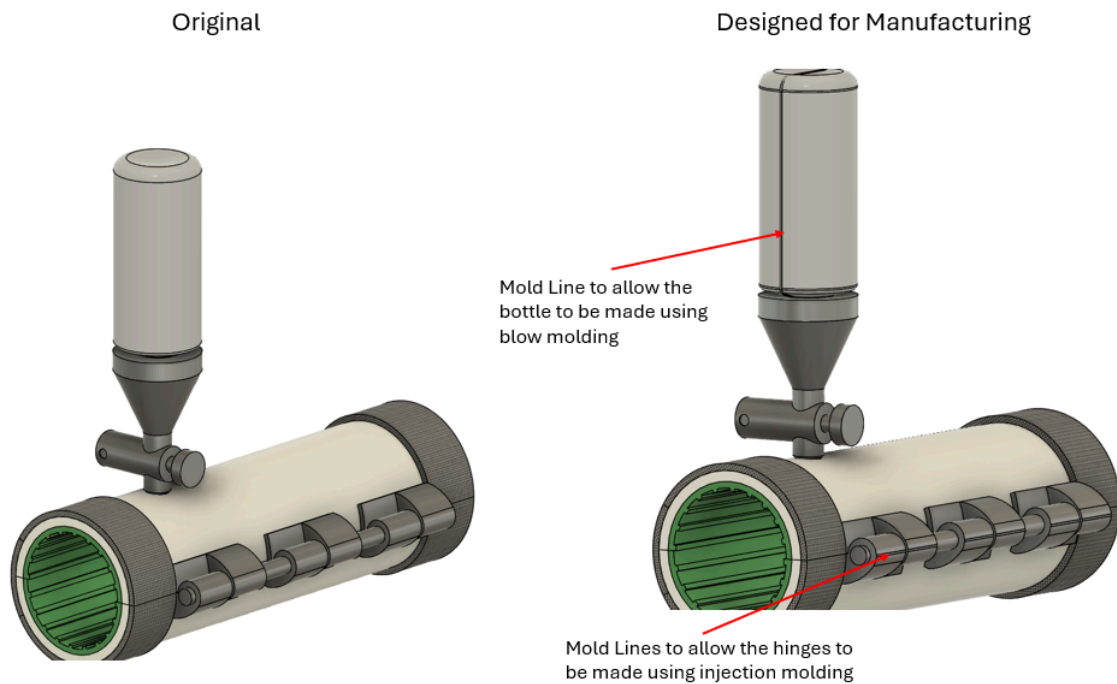
The PVC for the shell, velcro, scouring pads, rubber stoppers, and chemical bottle would all be purchased from a vendor then assembled assembly-line style. Each of these materials does not need much modification from us, the PVC just has to be cut in half with a hole drilled through one of the pieces, and the other components are attached with double sided tape or glue. Another reason why vendor purchasing would be effective for a larger assembly line would be the cost benefits of bulk ordering.

One of the struggles we could see with manufacturing our product would be with finding similar threading for our bottle and dispenser. To account for potential defects with combining an outside manufacturer's bottle to our custom dispenser, we could make our own bottle via blow molding.

Design Changes

We would change the material of hinges so that we can use injection molding or rotation molding. We also added mold lines to our CAD.

Adding a mold seam to the bottle since that is something that we would probably blow mold to get the right tolerances that we want.



DESIGN FOR ASSEMBLY

Reflection

One of the key advantages of our product is its low part count (9 parts, excluding the fasteners). This would make the assembly go very smoothly. Another advantage would be its modularity. There's the potential for multiple "sub-assemblies" to be combined with other assemblies (like one half of the PVC Shell or the dispenser). This improves the assembly process tremendously, as every piece is not entirely dependent on one another.

Some of the frustrations to our product's assembly would lie in faults/errors in the hinge placement or machining. If one of those hinges was misaligned, that would affect the ease of opening and closing the cleaner. As for manufacturing the product (specifically the stainless steel rod, and the PVC shell), there's room for human error in making sure the cuts are aligned with one another (if one or the other was too long, that wouldn't be the best quality of our product). The axles also had a tendency to fall out which we had to fix using hot glue on the ends of the axle, and this could also be fixed.

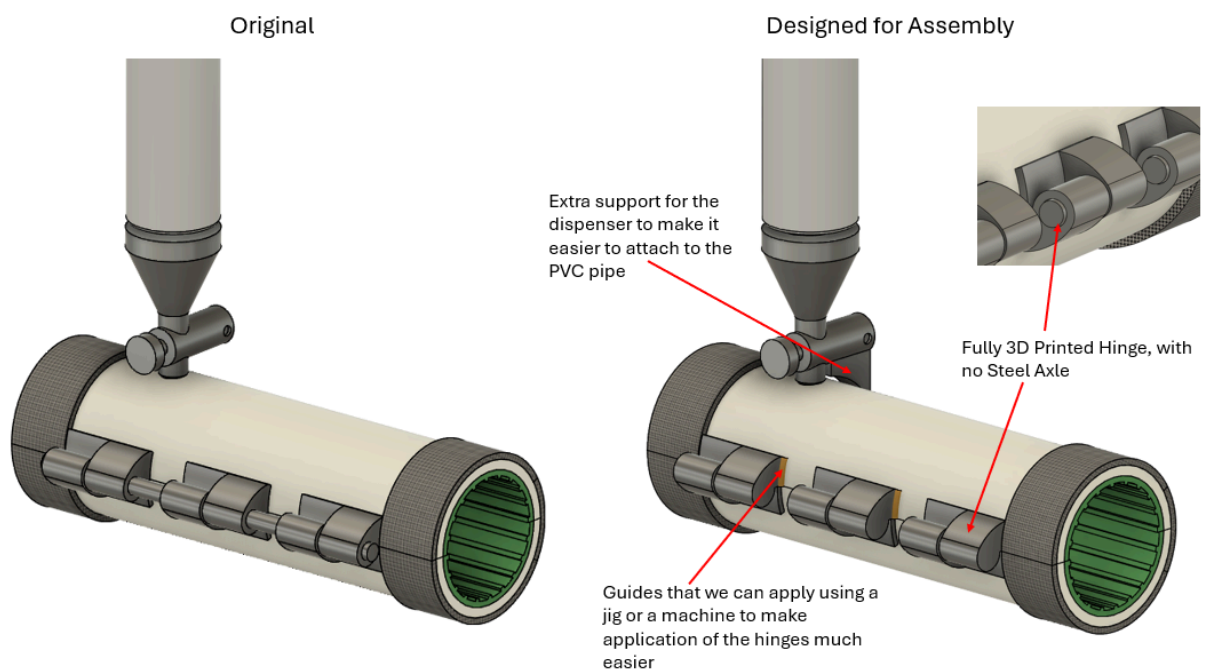
Another cause for concern would be the tolerancing of our 3D printed parts with the other fabricated part such as the PVC shell. For example a big issue that we had is that our 3D printed dispenser port was a little bit smaller than the hole that we could drill in the PVC, this caused it to be very loose and involved a lot of hot glue, VHB tape, and overall just a lot of issues in making that attach well.

Design Changes

One of the design changes we would make would be to add “guides” or make a jig to make attaching the hinges to the PVC pipe easier.

Another change would be to make our hinge fully 3D printed, and this would improve assembly by making it so that we did not need to add the steel axle which further reduces issues with the axles falling out or aligning everything well.

A third design change would be to add more material to the dispenser to allow more surface area to use double sided VHB tape to stick it very securely, even if the dispenser outer diameter is a bit smaller than the drilled one.



DESIGN FOR SUSTAINABILITY

Sustainability Reflection

When designing our product, we did not put sustainability at the forefront of our design choices. Although it seems like a simple product, there are many different materials that go into it making it more environmentally harmful. The materials we used are PVC pipe for the outer shell, Stainless Steel for the hinge rod, 3D printed PLA for our hinge brackets and custom-made acetone/liquid dispenser, and a commercially available polyester fiber internal scouring pads.

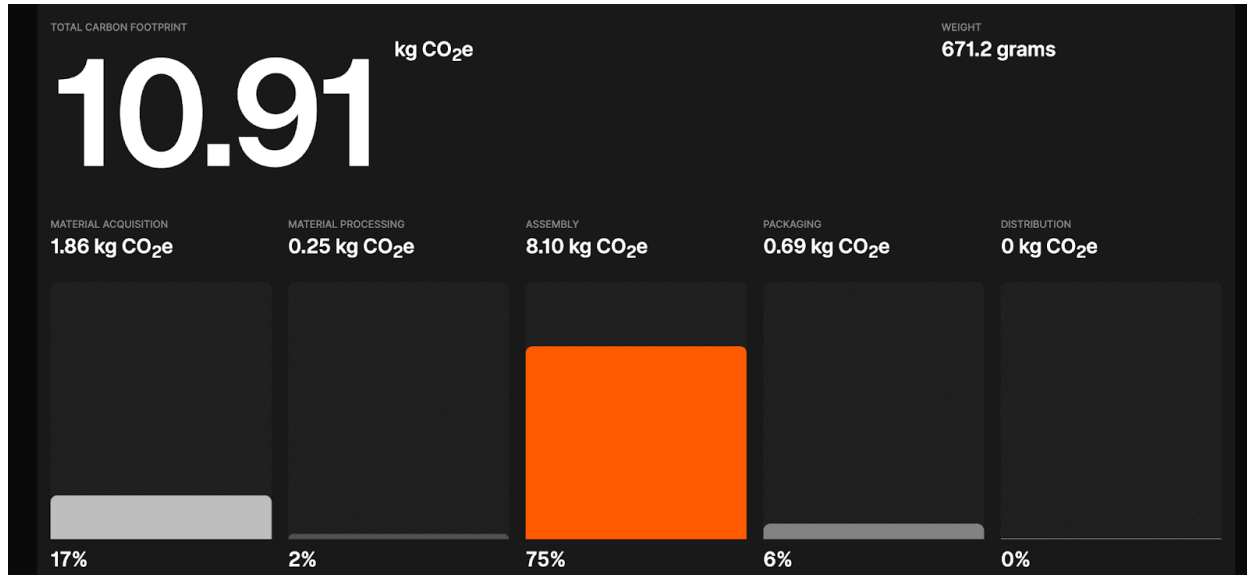
From these materials, we found that the HDPE and stainless steel are the most recyclable with HDPE being able to be recycled in almost any recycling container and stainless steel being

offered as a curbside pickup option. The PVC pipe section and 3D printed PLA parts are also recyclable, but they are often not accepted through just recycling bins and further steps need to be taken to drop it off at a recycling center. PLA is also biodegradable. The polyester fiber scouring pads are not that recyclable, and will have to be disposed of separately. This could pose issues in recycling since each of the parts are unable to be recycled together, and we would have to undo glue joints etc. to make it easy to recycle. One improvement could be to make the parts more easily removable.

The life cycle of our product would be pretty substantial with at least 5 years of use depending on how the user interacts with the product. Our scouring pads are somewhat difficult to remove and reattach due to strong tape attachment, and this would be the quickest wearing product which might have to be replaced every year.

Given what was said earlier, our product would most likely not be on its way to the recycling facility since most people would probably not take the time to separate each part separately. If there were changes to make the system easier to take our device apart, the user might be more inclined to do so and therefore making it eligible for recycling. We could also potentially make a recycling guide before launching our product that would help the user and recycling centers figure out which parts are recyclable and how to safely take apart the product.

2030 CALCULATOR RESULTS



Results from our 2030 Calculator Results

Looking at the results of the calculator, we can see that most of our carbon footprint is due to our assembly process. This makes sense because most of our components are coming from all parts of the country, and due to a variety of different materials it makes it more energy intensive. For

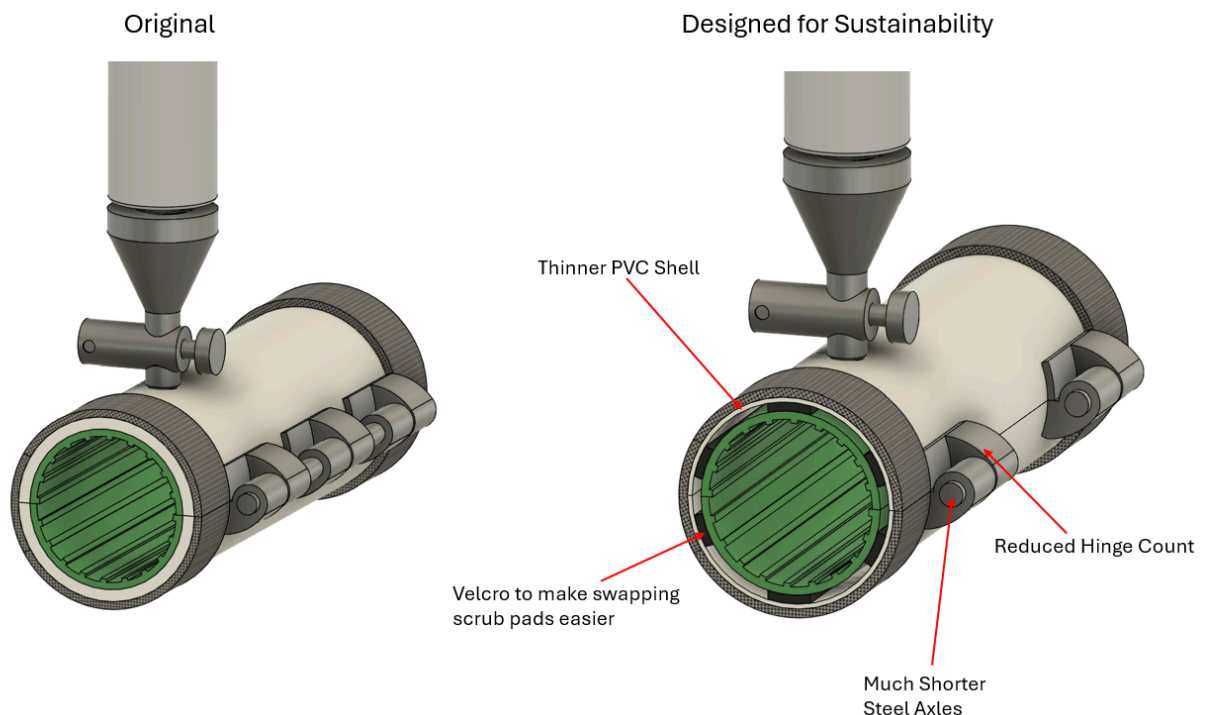
our calculations we found that most of the materials are shipped from Chicago and Ohio to a storage facility in Buffalo, NY and we set our company's location to Ithaca, NY so many products are also going from Buffalo to Ithaca. Another big part of this increased emission is that some of our parts such as the PVC pipe and the HDPE bottle already go through a lot of processing to get to us as a finished product that we later modify, this is also very energy intensive and creates more emission.

DESIGN MODIFICATIONS

One of the first design modifications would be to make modular scouring pads that you can easily replace over time. These could be fastened or hooked some way such as using hooks or velcro to make the pad easy to take out and put in, which would make our product last longer since the scouring pads are the quickest wearing part.

Another modification would be to reduce the number of hinges used or reduce the volume each one takes up. We definitely think that the mechanism could be achieved with fewer parts and fewer materials by cutting the axle sections down to only where the hinges are, and we could also use just 2 hinges.

A third modification would be to use a thinner wall thickness of PVC pipe since our current design has a pretty thick wall PVC pipe that is much stronger than it needs to be for our use case. Using thinner wall PVC would use less material and result in less emissions as well since it takes less energy to process.



Final Design Modifications

As we went through our design modifications we found that some of our changes somewhat interfered with one another, so we worked to synthesize some of our changes to be a cohesive whole. We chose to go with using 2 hinges instead of 3 due to our design for sustainability. We also made the hinges all plastic due to our design for assembly, and with mold lines for our design for manufacturing.

Next we chose to implement our bottle with mold lines as well to make it easy to blow mold, next we chose to not use the velcro between the PVC case due to the increased assembly and manufacturing challenge, and instead chose to use weaker double sided tape to make the scrub pads easier to replace and the tape easier to service. We still chose to go with the thinner PVC to reduce the material environmental cost for the design for sustainability. This change along with the larger inner diameter of the PVC also allows us to use a thicker scrub pad which should last longer. We also used the dispenser that had more surface area for attachment, to allow for greater ease of assembly.

FINAL PRODUCT RECOMMENDATION

The design's development cycle needs to be extended out further. Some small things will need to be improved before the design is ready to be produced on a larger scale and made practical enough for use in the job we intend, which is to clean pole vaults. The main items that need further development are the latching mechanism to close the device over a pole; something more permanent like hard ratchet bindings would be added to the product. Also, a new part needs to be developed to hold the rod for the hinges in place. Overall, making the design more permanent by using screws instead of adhesive and adhesive tape to hold it together would both make the design more manufacturable and more durable, allowing it to last longer. Durability is an important consideration for this product, as it will probably see a lot of abuse in the hands of athletes who use it to clean their sports equipment

Another option for the Pole Cleaner would be to close the project. We believe the market is too niche, and pole vaulters already have found methods for removing the tape, and are realistically unlikely to spend money on a product that will help them clean it faster.

REFLECTION

Throughout prototyping our pole-vault cleaner, our team navigated through the design process with teamwork and an iterative mindset. We first started with brainstorming as a team, and moved through quick cardboard prototypes and then from that selected which prototype we moved forward with. After selecting our main prototype, we only made brief modifications over

the initial prototype design we selected to arrive at our final product. A major success we had was our ability to adjust our new prototypes due to the problems of our old prototypes. For example, we replaced duct-tape hinges with more reinforced 3D printed hinges to address our durability concerns. We also adjusted our liquid dispensing mechanism to a 3D printed custom one as our original liquid dispensing mechanism did not dispense liquid at our desired rate. Our biggest problem with our cleaner was figuring out which material worked best to remove the gunk off the pole; we tested bristles and towels, but our final design included a scouring pad which worked the best. We found this primarily through iterative testing.

The best part for our team of the design process was the teamwork aspect. We divided up the work for the project aptly, and adhered to the strengths and weaknesses of our team members– the people familiar with CAD primarily did the CAD (with exceptions), and the people familiar with running a team assigned roles to each member. However, we also made each team member experience each part of the design process to increase the learning experience that comes with designing a new product, and to make sure the harder parts of the process do not get stuck to one individual. Overall, our team became more like a group of friends, where jokes can be thrown around and friendly banter was abundant, which furthered our ability to communicate, work more efficiently, and design the product to the best of our ability.

Rendering Package

<https://drive.google.com/drive/folders/1ilbhpXD57D-IPYnPru6Cs0L-S0O1QbQz?usp=sharing>

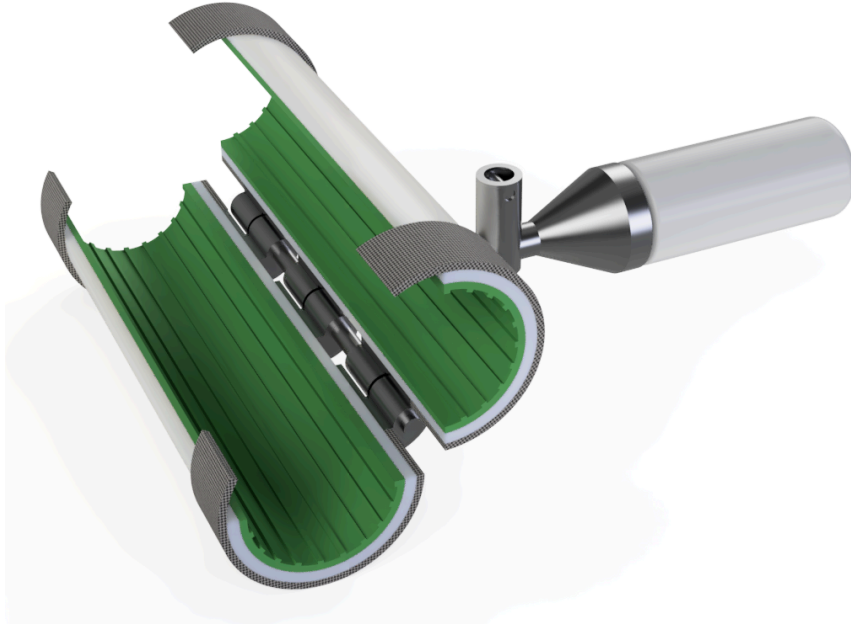
(High Resolution Files)



Our Real Life Built Prototype



Our Built Final Design in a Closed Position (Hinges and Dispenser are 3D Printed PLA, they appear more Matte in Real Life than what we could achieve in our Rendering, the color is both Gray in the Rendering and our Real Life Model)



Our Built Final Design in an Open Position



Version 2 Design with A Metal Case instead of PVC



Version 2 Design with a Metal Case in an Open Configuration



Version 3 Design with a Metal Case and the Scrub Pad Changed to Brown



Version 4 Design with a Blue Scrub Pad, Deep Black Velcro and Metal Case



Version 4 Design in an Open Configuration



Version 5 Design with a Metal Case, White Velcro, and a Blue Scrub Pad



Version 5 Design in an Open Configuration



Version 6 Design with a Metal Case, White Velcro, Blue Scrub Pad, and Black Bottle



Version 7 Design with Deep Black Velcro, a Metal Case, Blue Scrub Pad, and a Black Bottle